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Dave's Daily

AI • Bayesian optimization • US equities

tldr

1. Meta released Muse Wonder, a frontier model meant to act for a person and remember long conversations. On Meta's own memory test the scores are 85.6 and 85.5, against 81.4 for GPT-6 Astra. The consumer agent gets the model first for subscribers.
2. JPMorgan and Goldman underwrote $1.3 billion of Volta's $5 billion AI loan. The $3.7 billion piece that buys about 36,000 Nvidia chips is not underwritten. The yield is about 11%, and the debt is unrated.
3. A new database tuner treats the manual's knob orderings as the search space. On PostgreSQL, about 35% of ordinary trials crashed the server. With those rules built in, search reached the old best result up to 12.5 times faster.

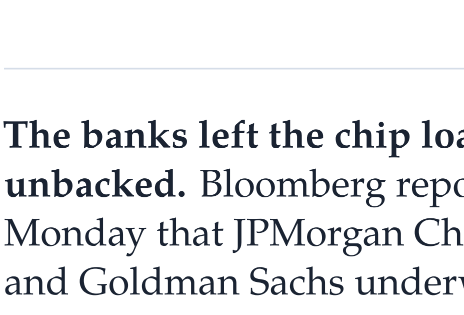
AI

A frontier model that is supposed to act. On Monday Meta released Muse Wonder and Muse Wonder Full (Meta; TechCrunch). Wonder is on the Meta Model API and in Muse Code. Wonder Full is only for Muse Code subscribers. The consumer agent, Muse, will get Wonder later, and subscribers go first (Meta research). TechCrunch reports an API price of $4 per million input tokens and $20 per million output tokens. The model card lists a context of one million tokens and a release date of 12 October 2026.

The memory claim is Meta's own test. Meta calls the feature Active Memory: carry what matters across conversations and bring it back when a task needs it. The card's benchmark, MemoryBench, uses conversations of 64,000 to 100,000 tokens, with distractors and facts that later change. An automated judge scores the answers. Wonder scores 85.6 and Wonder Full 85.5. Claude Fable 5.1 scores 79.0, GPT-6 Astra 81.4, and Claude Opus 5.5 81.5 (chart). The smaller model wins this row. The test is internal. The card does not include an outside lab's score for it.

What a person hands over. The research post shows the model planning a run from someone's preferences, ordering gifts from an Instagram profile, and working through a lease or an insurance claim. A personal agent of that kind reads messages, calendars, and documents. Meta says a fuller safety report is still coming. The card says the models are not intended for anyone under 18. With mitigations on, Meta rates residual risk in cybersecurity, chemical and biological misuse, and loss of control as moderate or lower. Without those mitigations, the card says Meta does not rule out a severe rating in the chemical and biological domain. The Wall Street Journal says global downloads of the Muse agent have slowed, so the model arrives while the product is trying to hold users.

Hypothesis: a lead of about four points on an internal memory test is a real gap on that test. It is still Meta's measurement of Meta's task.



MemoryBench, Meta's internal test of recall across conversations of 64,000 to 100,000 tokens. Higher is better. The axis starts at zero. Source: Meta, Muse Wonder model card, 12 October 2026.

Markets

The banks left the chip loan unbacked. Bloomberg reported Monday that JPMorgan Chase and Goldman Sachs underwrote a $1.3 billion term loan for Volta Infrastructure Holdings and left a $3.7 billion chip loan without an underwrite (Mint, carrying Bloomberg). People familiar with the deal described the split as a limit on the banks' risk. An underwrite usually covers the whole package, so the borrower still gets the money if investors walk away. Here the larger piece has to find buyers on its own. Commitments are due 14 October. Both banks and Volta declined to comment.

The price, and who it shuts out. The debt is pitched at a yield of about 11%, which Bloomberg calls one of the highest in the leveraged-loan market now used for AI. The loans are unrated, so collateralized loan obligations, the usual buyers, typically cannot take them. Friday's price talk was 6.25 to 6.50 percentage points over the benchmark, at 97 to 98 cents (PitchBook).

A neocloud, a miner, and a lab. Volta was founded this year by Ricard Boada and Sofia Gumuzio, former Brookfield infrastructure executives. It rents chips. Bloomberg has previously reported that the loans fund a data-center complex in Norway run with Bitdeer, a bitcoin miner, and that Anthropic has a six-year agreement for capacity there. The $1.3 billion loan pays that lease. The $3.7 billion loan buys about 36,000 Nvidia GPUs. Earlier loans for CoreWeave, Lambda, and Crusoe cleared at lower yields, Bloomberg says. These notes describe the structure. They are not a forecast of share prices.

Bayesian optimization

Scaling: search only where the server can boot. Fangping Lan, Qi Zhang, and Eduard Dragut, at Temple, tune a database by Bayesian optimization inside the region the manual already marks as valid (arXiv:2610.09276, 7 October; code). PostgreSQL 13 exposes 263 knobs. A preliminary run on TPC-C, with 46 knobs and the documented ranges, crashed the server on about 35% of trials. One written rule says the reserved superuser connections must be fewer than the maximum connections. Standard tuners treat the box of ranges as the search space and learn those rules by sampling crashes.

The mechanism. Ordering rules, such as a minimum write-ahead-log size no larger than the maximum, become the domain. The knobs form a graph. CATune samples a root knob first, then each child inside the interval its parents still allow, so every candidate is feasible when it is drawn. For two of their examples, the chance a random draw satisfies both orderings is about one in a million. On a 10-knob example, one ordering removed about half the box.

The experiment. Under default ranges, CATune reached the result SMAC had after 200 iterations 12.5 times sooner on TPC-C and 2.27 times sooner on TPC-H. Final throughput was 33.39% higher on TPC-C. Under the narrower ranges experts suggest, the TPC-C throughput gain was 63.37%, and the speedup to the old best was 5.41 times.

Where the language model enters. The constraints come from the manual, and the authors ask a language model to extract them. Direct prompting produced dependencies the text did not support, including some pointed the wrong way. Two of those false constraints measurably slowed the search. Their pipeline asks for a span of evidence, reflects, and uses a second model as a judge before a constraint is kept. Hypothesis: most of the speed comes from the smaller feasible set, once a second check throws out constraints the language model invented. The speedups are against SMAC on PostgreSQL, for TPC-C and TPC-H.

Updated 12 October 2026. Phone and letter match. Sources: Meta, 12 October; Bloomberg via Mint, 19:01 UTC; arXiv, 7 October. The loan note is not a forecast. Created with malvin.